

ARMA: Estimation

Yule-Walker Equations

Consider AR(p) model: $x_{n+m} = \phi_1 x_{n+m-1} + \cdots + \phi_p x_{n+m-p} + w_{n+m}$

$$\begin{bmatrix} \gamma(0) & \gamma(1) & \cdots & \gamma(p-1) \\ \gamma(1) & \gamma(0) & \cdots & \gamma(p-2) \\ \gamma(2) & \gamma(1) & \cdots & \gamma(p-3) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(p-1) & \gamma(p-2) & \cdots & \gamma(0) \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \\ \phi_3 \\ \vdots \\ \phi_p \end{bmatrix} = \begin{bmatrix} \gamma(p) \\ \gamma(p-1) \\ \gamma(p-2) \\ \vdots \\ \gamma(1) \end{bmatrix}$$

$$\mathbf{\Gamma}_p \boldsymbol{\phi} = \boldsymbol{\gamma}_p$$

$$\sigma_w^2 = \gamma(0) - \boldsymbol{\phi}' \boldsymbol{\gamma}_p$$

If $\gamma(h)$ is known, we can find ϕ, σ_w^2 .

Yule-Walker Estimator

$$\hat{\gamma}(h) = \frac{1}{n} \sum_{t=1}^{n-h} (x_{t+h} - \bar{x})(x_t - \bar{x}), \quad h = 0, 1, \dots, n-1$$

$$\bar{x} = \frac{1}{n} \sum_{t=1}^n x_t$$

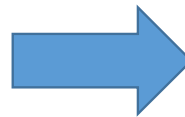
$$\hat{\gamma}(-h) = \hat{\gamma}(h)$$

$$\mathbf{R}_p = \begin{bmatrix} 1 & \rho(1) & \rho(2) & \dots & \rho(p-1) \\ \rho(1) & 1 & \rho(1) & \dots & \rho(p-2) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho(p-1) & \rho(p-2) & \rho(p-3) & \dots & 1 \end{bmatrix}$$

$$\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\Gamma}}_p^{-1} \hat{\boldsymbol{\gamma}}_p$$

$$\hat{\sigma}_w^2 = \hat{\gamma}(0) - \hat{\boldsymbol{\phi}}' \hat{\boldsymbol{\gamma}}_p$$

Write in term of $\rho(h)$



$$\hat{\boldsymbol{\phi}} = \hat{\mathbf{R}}_p^{-1} \hat{\boldsymbol{\rho}}_p$$

$$\hat{\sigma}_w^2 = \hat{\gamma}(0)(1 - \hat{\boldsymbol{\phi}}' \hat{\boldsymbol{\rho}}_p)$$

Asymptotic property

For $n \rightarrow \infty$,

$$\sqrt{n}(\hat{\phi} - \phi) \xrightarrow{d} N(0, \sigma_w^2 \mathbf{\Gamma}_p^{-1}), \quad \hat{\sigma}_w^2 \xrightarrow{p} \sigma_w^2$$

$$\sqrt{n}\hat{\phi}_{hh} \xrightarrow{d} N(0,1) \text{ for } h > p$$

MA and ARMA is AR(p), $p \rightarrow \infty$.

Use maximum likelihood model.

ARIMA(p, d, q)

A process x_t is said to be **ARIMA**(p, d, q) if $\nabla^d x_t = (1 - B)^d x_t$ is ARMA(p, q).

$$\phi(B)(1 - B)^d x_t = \theta(B)w_t$$

If $E(\nabla^d x_t) = \mu$, the model is

$$\phi(B)(1 - B)^d x_t = \delta + \theta(B)w_t,$$

where $\delta = \mu(1 - \phi_1 - \phi_2 - \dots - \phi_p)$.

Seasonal ARIMA (SARIMA)

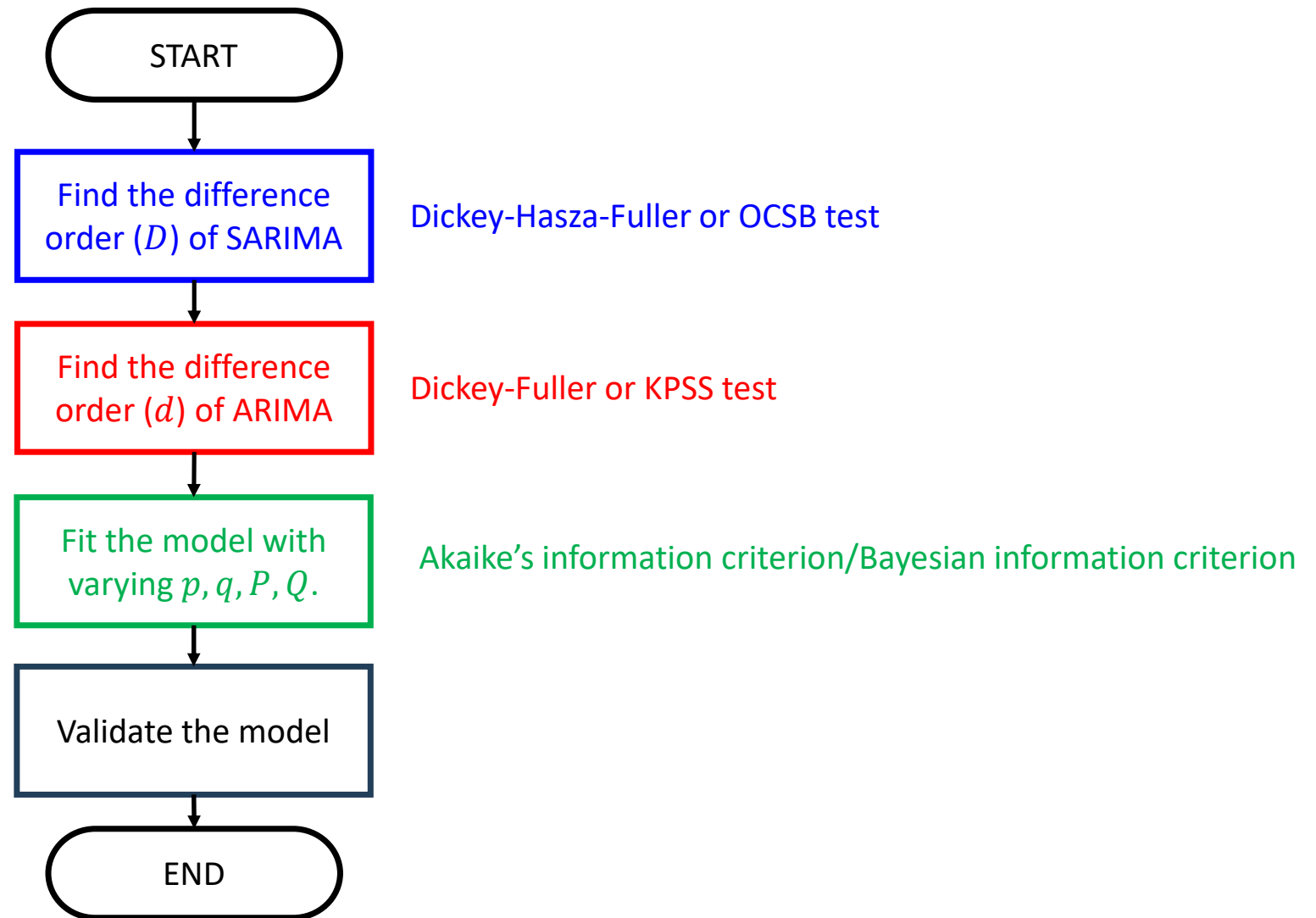
$$\text{ARIMA}(p,d,q): \quad \phi(B)\nabla^d x_t = \theta(B)w_t$$

$$\text{SARIMA}(p,d,q) \times (P,D,Q):$$

$$\Phi_P(B^S)\phi(B)\nabla_S^D\nabla^d x_t = \delta + \Theta_Q(B^S)\theta(B)w_t$$

$$\begin{aligned} (1 - \Phi_1 B^S - \Phi_2 B^{2S} - \dots - \Phi_P B^{PS})\phi(B)(1 - B^S)^D\nabla^d x_t \\ = \delta + (1 + \Theta_1 B^S + \Theta_2 B^{2S} + \dots + \Theta_Q B^{QS})\theta(B)w_t \end{aligned}$$

Flowchart



Dickey-Fuller test (DF test)

Consider AR(1) and ARIMA(0,1,0) processes.

$$\text{AR(1) process: } x_t = \phi_1 x_{t-1} + w_t$$

$$\text{ARIMA(0,1,0) process: } x_t = x_{t-1} + w_t \rightarrow \phi_1 = 1$$

$$x_t - x_{t-1} = (\phi_1 - 1)x_{t-1} + w_t$$

$$\nabla x_t = \alpha x_{t-1} + w_t$$

$$H_0: \alpha = 0 \rightarrow \text{ARIMA(0,1,0)}$$

$$H_1: \alpha < 0 \rightarrow \text{ARMA(1).}$$

Augmented DF test (ADF test)

- Assume that the order of AR process is p .
User defined parameter

KPSS test

Stationary: $|\text{root}| > 1$

root = 1

- Kwiatkowski-Phillips-Schmidt-Shin test

$$\phi(B)(1-B)^d x_t = \delta + \theta(B)w_t$$

$$\Phi(B)x_t = \delta + \theta(B)w_t$$

$H_0: \Phi(B)$ contains unit root $\longrightarrow$ Take difference

$H_1: \Phi(B)$ contains no unit root

Dickey-Hasza-Fuller test (DHF test)

Consider ARMA(m) and SARIMA(0,0,0)(0,1,0, S) processes.

ARMA process: $x_t = \phi_1 x_{t-S} + w_t$

SARIMA process: $x_t = x_{t-S} + w_t \rightarrow \phi_1 = 1$

$$x_t - x_{t-S} = (\phi_1 - 1)x_{t-1} + w_t$$

$$\nabla^S x_t = \alpha x_{t-1} + w_t$$

$H_0: \alpha = 0 \rightarrow \text{SARIMA process}$

$H_1: \alpha < 0 \rightarrow \text{ARMA process.}$

OCSB test

Stationary: $|\text{root}| > 1$

AR root = 1

- Osborn-Chui-Smith-Birchenhall test

$$\underbrace{\Phi_P(B^s)\phi(B)}_{\Psi(B)} \nabla_s^D \nabla^1 x_t = \delta + \Theta_Q(B^s)\theta(B)w_t$$

$$\Psi(B)y_t = \delta + \Theta_Q(B^s)\theta(B)w_t$$

$H_0: \Psi(B)$ contains unit root

$H_1: \Psi(B)$ contains no unit root

Fitness criterion

- Akaike's information criterion (AIC)

$$\text{AIC} = -\text{error} + \text{number of parameters}$$

- Bayesian information criterion (BIC)

$$\text{BIC} = -\text{error} + \ln(\text{sample size}) \cdot \text{number of parameters}$$

└ Penalty of complex model is higher.

Model validation

$\frac{x_t - \hat{x}_t^{t-1}}{\sqrt{\hat{P}_t^{t-1}}}$ is iid process with zero mean and variance of one.

- Time plot of $\frac{x_t - \hat{x}_t^{t-1}}{\sqrt{\hat{P}_t^{t-1}}}$.
- Q-Q Plot
- ACF and PACF of $x_t - \hat{x}_t^{t-1}$.
- Ljung-Box statistic `from statsmodels.stats.diagnostic import acorr_ljungbox`